

prob

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Problems On Strings

```
In [2]:
import math
s = "james"

midIdx=math.floor(len(s)/2)
string = s[0]+s[midIdx]+s[-1]

print(string)
```

```
Out [2]:
jms
```

```
In [3]:
s2 = s[midIdx-1]+s[midIdx]+s[midIdx+1]
print(s2)
```

```
Out [3]:
ame
```

```
In [4]:
s3 ="hello world".title()
print(s3)
```

```
Out [4]:
Hello World
```

```
In [5]:
sam = "America"
sjap = "Japan"

def findmididx(stringparameter):
    return math.floor(len(stringparameter)/2)

print(sam[0]+sjap[0]+sam[findmididx(sam)]+sjap[findmididx(sjap)]+sam[-1]+sjap[-1])
```

```
Out [5]:
AJrpan
```

```
In [6]:
slinput = list(input().split())
print(slinput)
```

```
Out [6]:
['america', 'russia', 'indonesia', 'japan']
```

```
In [7]:
```

```
res = ""
for x in slinput:
    res += x[0]
for x in slinput:
    res += x[findmididx(x)]
for x in slinput:
    res += x[-1]
print(res)
```

Out [7]:
arijrsnpaaan

```
In [8]:
s = "12321"
print(s[::-1])
```

Out [8]:
12321

```
In [9]:
s = "1234321"

if s == s[::-1]: print("Palindrome")
else: print("Not palindrome")
```

Out [9]:
Palindrome

```
In [10]:
c = 0
word = "aeiouaeiou".lower()

vowel = ['a' , 'e' , 'i' , 'o' , 'u']
for i in word:
    if i in vowel:
        c+=1
print(c)
```

Out [10]:
10

```
In [11]:
emails = ['pamarthisiva144@gmail.com' , 'nagu@yahoo.com']
```

```
for x in emails:
    idx_ano = x.index('@')
    idx_dot = x.index('.')
    print(x[idx_ano+1:idx_dot:])
```

Out [11]:
gmail
yahoo

```
In [12]:
s = 'Hello World From Python'
r = s.split()
r = r[::-1]
res=''
for i in r:
    res+=i+" "
print(res)
```

```
Out [12]:  
Python From World Hello
```

```
In [17]:  
s = 'Hello World From Python'  
l = s.split()  
  
res = ''  
for x in l :  
    res += x[::-1]+" "  
print(res)
```

```
Out [17]:  
olleH dlroW morF nohtyP
```

```
In [14]:  
s = 'Hello World From Python'  
r = s.split()  
r = r[::-1]  
res=[]  
for i in r:  
    res.append(i)  
  
ans = ' '  
for x in res:  
    ans += x[::-1]+" "  
print(ans)
```

```
Out [14]:  
nohtyP morF dlroW olleH
```

```
In [33]:  
a = "listen".lower()  
b = "silent".lower()  
  
print("Anagrams" if sorted(a) == sorted(b) else "Not anagrams")
```

```
Out [33]:  
Anagrams
```

Map

```
In [36]:  
def mul(x):    return x*2  
l = [1,2,3]  
res = list(map(mul , l))  
print(res)
```

```
Out [36]:  
[2, 4, 6]
```

```
In [37]:  
nums = [1,2,3,4]  
sqnums = list(map(lambda x : x**2 , nums ))  
  
print(sqnums)
```

```
Out [37]:  
[1, 4, 9, 16]
```

```
In [40]:  
rate = [70 , 66, 50 , 40]  
currency = [200 , 100, 50 , 20]
```

```
res = list(map(lambda x,y : x*y , rate , currency))
print(res)
```

```
Out [40]:
[14000, 6600, 2500, 800]
```

```
In [43]:
metal = ["gold" , "silver"]
res = list(map(str.capitalize, metal))
print(res)
```

```
Out [43]:
['Gold', 'Silver']
```

```
In [51]:
res = list(map(lambda x:x[0] , metal))
r = "".join(res)
print(r)
```

```
Out [51]:
gs
```

Filter

```
In [53]:
def even(x):
    if x%2 == 0 : return True
    else : return False
```

```
l = [1,2,3,4,5,6]
res = list(filter(even , l))
print(res)
```

```
Out [53]:
[2, 4, 6]
```

```
In [55]:
age = [25 , 65 , 89 , 45 , 23]

res = list(filter(lambda x: x>25 , age))
print(res)
```

```
Out [55]:
[65, 89, 45]
```

```
In [58]:
data = [0 , "hello" , 42 , None , "world" , "false" , False , 42]

falsywords = [0 , "" , False , None]

print(list(filter(lambda x: True if x not in falsywords else False , data )))
```

```
Out [58]:
['hello', 42, 'world', 'false', 42]
```

Reduce

```
In [59]:
from functools import reduce as red
```

```
In [60]:
```

```
l = [10 , 20 , 30 , 40 ]
```

```
tot = red(lambda x,y :x+y , l)  
print(tot)
```

Out [60]:

100

In [62]:

```
l = [1,2,3,4,5]  
max1 = red(lambda x,y:x if x>y else y , l)  
print(max1)
```

Out [62]:

5